

Sphere and Schur sources for the localized Weil form

Exact local norms and the inverse bulk matching problem

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Abstract

We study the problem of realizing the complete localized Weil quadratic form as the norm of a linearly prepared state in an independently positive quantum or boundary system. The intended difficulty is an exact matching identity, potentially conditional on construction of a specified field theory. We give explicit calculations in sphere and Schur quantization. The real Gaussian boundary module supplies the quarter-shift gamma factor and the positive oscillator tower, while its natural dilation pairing differs from the Weil form. In an elementary Schur representation, an electric dressing followed by one magnetic insertion realizes the full positive single-prime reference, including every negative prime-power coefficient and its compulsory positive contact. A single quantization parameter works for all primes, and the insertion is justified by a graph-domain approximation. We derive restrictions on higher magnetic powers and exclude a constant-vector coherent gluing of two primes through its unwanted mixed translation atoms. Further exact tests cover interacting sphere states, a conformal electric sector, gauge transfer, and finite rational feedback. The complete contact and signed pole matching remain open. These results select source constructions; they do not establish Weil positivity or a field-theoretic reduction of the Riemann hypothesis.

1 Introduction

The Weil criterion expresses the Riemann hypothesis as positivity of an explicit quadratic form on compactly supported test functions. An inverse bulk approach asks for a quantum system, a boundary or defect preparation, and an independently positive physical pairing whose pullback is exactly that form. Interactions are compatible with this objective: a preparation linear in the input produces a quadratic state norm even when the underlying dynamics is interacting. What must be derived is the entire pairing, with its physical adjoint and normalization.

Here “boundary” means the locus or observable sector through which the arithmetic input enters the system. It may be a spatial boundary, a line defect, an interface, a gluing cut, or a module for a protected operator algebra. This flexibility permits the use of sphere and Schur quantization, where explicit representations and positive pairings are available [5, 6, 7]. Conformal symmetry can constrain a chosen defect, but does not identify arbitrary deformations or source maps. Pure four-dimensional Yang–Mills is also a possible long-term construction target; no identification with it is made in this paper.

1.1 Principal result and its limitation

The most concrete arithmetic result concerns the elementary Schur representation. For a fixed quantization parameter $0 < q < 1$, we construct an electric dressing and one magnetic insertion whose norm on Laurent polynomials h is

$$\|\Phi_{q,r,d}[h]\|^2 = \int_{S^1} |h(z)|^2 \frac{dr(1+r)}{1-r} \frac{|1-z|^2}{|1-rz|^2} \frac{d\theta}{2\pi}, \quad z = e^{i\theta}, \quad 0 < r < 1, \quad (1.1)$$

with $d > 0$. Its Fourier coefficients are

$$\frac{dr(1+r)}{1-r} \frac{|1-z|^2}{|1-rz|^2} = \frac{2dr}{1-r} - d \sum_{n \geq 1} r^n (z^n + z^{-n}). \quad (1.2)$$

Thus $r = p^{-1/2}$ and $d = \log p$ give every required single-prime repetition coefficient. The arithmetic parameters enter the preparation; q is held fixed. Theorem 6.2 proves this identity and the domain statement needed for its magnetic insertion. An explicit unitary decomposition then relates the circle calculation to the real logarithmic coordinate of the Weil form.

The positive contact $2dr/(1-r)$ is part of the result. This is a Schur realization of the positive prime reference already present in the project's conservative return construction. Neither the reference form nor the geometric Fourier identity is claimed as new. The additional content is the specified operator preparation, its source normalization, its graph-domain justification, and its compatibility with one fixed Schur parameter. A physical interface implementing the continuous-input lift has not been constructed.

1.2 Other results and the remaining question

The Gaussian real boundary module identifies the archimedean quarter-shift and oscillator levels, but its natural state-smearing norm has a different spectral weight. In the interacting $T[SU(2)]$ sphere model we evaluate an infinite family of charged-state norms and show that bare powers do not have geometric return ratios. A natural electric preparation in the conformal $SU(2)$ theory with four hypermultiplets also fails, as does the undressed elementary Schur return prescription. These are exclusions of specified preparations, not of the theories.

Two further results govern the new dressed source. Its finite magnetic domain does not permit unrestricted use of higher-power identities. Moreover, gluing the prime filters through fixed output vectors produces an atom at $\log(3/2)$ unless those vectors are orthogonal (Proposition 7.2). Orthogonality restores the independent positive contacts. A successful joint construction must therefore change the preparation or the gluing operation in a substantive way.

The background and normalization are developed in Section 2, independently of any companion manuscript. Section 3 states the conditional construction-to-RH implication and a general source-regularity restriction. Sections 4–7 contain the main calculations. Section 8 formulates the next complete fitting problem. The appendices retain positive gamma refinement and earlier gauge and rational-feedback tests, with proofs sufficient to use them without consulting the research notes.

The classical Weil criterion, gamma identities, and established representation formulas are explicitly attributed. The comparison calculations are working results, not a claim of literature priority. In particular, an exact identity between local phase derivatives and the prime terms is not a positivity theorem. No candidate in this draft matches the complete first-prime Weil form, and no all-support result follows from the local norm in (1.1).

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2 The complete Weil form in logarithmic coordinates

2.1 Conventions and the classical criterion

All Hilbert inner products are conjugate-linear in the first argument. For $L > 0$, let $I_L = (-L/2, L/2)$ and let E_L denote extension by zero from I_L to $\mathbb{R}$. Unless another domain is specified, f belongs to $C_c^\infty(I_L; \mathbb{C})$ and $F = E_L f$. Our Fourier and translation conventions are

$$\widehat{F}(\tau) = \int_{\mathbb{R}} e^{-i\tau x} F(x) dx, \quad U_d F(x) = F(x - d), \quad \|F\|^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{F}(\tau)|^2 d\tau. \quad (2.1)$$

Write $\widetilde{F}(x) = \overline{F(-x)}$ and $g = F * \widetilde{F}$. Then $g(0) = \|F\|^2$, $g(-r) = \overline{g(r)}$, and $g(-r) = \langle F, U_r F \rangle$. In particular, the correlations vanish for $|r| \geq L$.

For $r > 0$ set

$$n_\gamma(r) = \frac{e^{-r/2}}{1 - e^{-2r}} = \sum_{n \geq 0} e^{-a_n r}, \quad a_n = 2n + \frac{1}{2}. \quad (2.2)$$

One normalization of the additive Weil functional is

$$\begin{aligned} \mathcal{W}(g) &= \int_{\mathbb{R}} 2 \cosh(x/2) g(x) dx - \sum_{p, m \geq 1} (\log p) p^{-m/2} \{g(m \log p) + g(-m \log p)\} \\ &\quad - (\log(4\pi) + \gamma_E) g(0) \\ &\quad - \int_0^\infty [g(r) + g(-r) - 2e^{-r/2} g(0)] n_\gamma(r) dr. \end{aligned} \quad (2.3)$$

Here p runs over rational primes and γ_E is Euler's constant. The integrand at the origin has a finite limit, and the prime sum is finite for compact support. This normalization can be found in Suzuki's account of the Weil form [1, Section 1.1]; for its geometric and trace interpretation see [2].

The Riemann hypothesis (RH) asserts that every nontrivial zero of the meromorphic continuation of $\zeta(s) = \sum_{n \geq 1} n^{-s}$ has real part $1/2$. We use the following classical theorem as the number-theoretic input; we do not reprove the explicit formula or the Weil criterion.

Theorem 2.1 (Classical Weil positivity criterion). *RH holds if and only if $\mathcal{W}(F * \widetilde{F}) \geq 0$ for every complex $F \in C_c^\infty(\mathbb{R})$.*

Every compact support is contained in some I_L . Consequently it is sufficient, and necessary, to establish this positivity for every $L > 0$ and every test function in I_L .

2.2 Positive gamma energy and the fixed contact

Define

$$\begin{aligned} K[F] &= \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}} |F(x) - F(y)|^2 n_\gamma(|x - y|) dx dy \\ &= \int_0^\infty [2g(0) - g(r) - g(-r)] n_\gamma(r) dr. \end{aligned} \quad (2.4)$$

This is nonnegative without any arithmetic hypothesis. Expanding (2.2) and applying Plancherel gives

$$K[F] = \frac{1}{2\pi} \int_{\mathbb{R}} b(\tau^2) |\widehat{F}(\tau)|^2 d\tau, \quad (2.5)$$

$$b(\tau^2) = \sum_{n \geq 0} \frac{2}{a_n} \frac{\tau^2}{a_n^2 + \tau^2} = \operatorname{Re} \psi\left(\frac{1}{4} + i\tau/2\right) - \psi\left(\frac{1}{4}\right), \quad (2.6)$$

where $\psi = \Gamma'/\Gamma$. Indeed, $2 \int_0^\infty (1 - \cos(\tau r))e^{-ar} dr = (2/a)\tau^2/(a^2 + \tau^2)$; nonnegativity justifies interchange with the sum. The second identity in (2.6) is the digamma partial-fraction expansion [3].

The archimedean part of (2.3) equals $K[F] + w_0\|F\|^2$, where

$$\begin{aligned} w_0 &= -\log(4\pi) - \gamma_E - 2 \int_0^\infty (1 - e^{-r/2})n_\gamma(r) dr \\ &= \psi(\tfrac{1}{4}) - \log \pi = -\gamma_E - \frac{\pi}{2} - 3 \log 2 - \log \pi < 0. \end{aligned} \quad (2.7)$$

To verify the constant, the integral in the first line is $\sum_{n \geq 0} (a_n^{-1} - (2n+1)^{-1}) = \frac{1}{2}[\psi(1/2) - \psi(1/4)]$. The constant is therefore fixed by the arithmetic normalization. It is not a parameter available to adjust a positive source.

2.3 The signed poles and the localized target

The first term of (2.3) has kernel $2 \cosh((x-y)/2)$. Using the hyperbolic addition formula gives

$$P_L[f] = 2 \left| \int_{I_L} f(x) \cosh(x/2) dx \right|^2 - 2 \left| \int_{I_L} f(x) \sinh(x/2) dx \right|^2. \quad (2.8)$$

It is a rank-two Hermitian form, with one positive and one negative direction for $L > 0$. Its pointwise positive kernel does not make it a positive quadratic form.

The exact target for this paper is

$$\begin{aligned} Q_L[f] := \mathcal{W}(F * \tilde{F}) &= K[F] + w_0\|f\|^2 + P_L[f] \\ &\quad - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \operatorname{Re} \langle F, U_{m \log p} F \rangle. \end{aligned} \quad (2.9)$$

Its polarization is denoted by $Q_L(f, g)$. At equality $m \log p = L$ the compressed correlation is zero. The coefficient is the primitive length $\log p$, not the repeated length $m \log p$.

On a fixed interval the prime and pole terms define bounded operators. The gamma multiplier satisfies

$$b(\tau^2) = \log |\tau| - \log 2 - \psi(\tfrac{1}{4}) + O(|\tau|^{-2}) \quad (|\tau| \rightarrow \infty). \quad (2.10)$$

The natural energy domain is

$$\mathcal{D}_{\gamma, L} = \left\{ f \in L^2(I_L) : \int_{\mathbb{R}} \log(2 + |\tau|) |\widehat{E_L f}(\tau)|^2 d\tau < \infty \right\}. \quad (2.11)$$

The positive gamma energy is closed on this domain; it is the restriction of the closed nonnegative Fourier-multiplier form to the closed subspace of functions supported in $\bar{I}_L$. The bounded other terms give a closed semibounded form on the same domain. All matching statements below are first required on smooth compactly supported inputs. The Weil criterion itself needs only that test space.

Away from the diagonal the distribution kernel of K is $-n_\gamma(|x-y|)$, which is smooth whenever $x \neq y$. The prime contribution instead has atoms at $x-y = \pm m \log p$. This distinction will provide exact tests for candidate preparations.

Three small intervals already distinguish several issues:

L	Active prime powers	Test
1	2	First contact, pole and translation identity
5/4	2, 3	Joint source and mixed returns
3/2	2, 3, 4	First nontrivial repetition of 2

A calculation on any one interval is preliminary. It does not establish the all-support statement of Theorem 2.1.

3 What an independently positive realization requires

3.1 State norms, constructive hypotheses, and linear sources

Definition 3.1 (Exact positive realization). An exact positive realization of the Weil form consists of a specified model M with a positive Hilbert space $\mathcal{H}_M$ and complex-linear maps $\Phi_{M,L} : C_c^\infty(I_L) \rightarrow \mathcal{H}_M$ such that

$$\langle \Phi_{M,L}(f), \Phi_{M,L}(g) \rangle = Q_L(f, g) \quad (L > 0). \quad (3.1)$$

The model, its physical inner product, source rule and normalization must be specified independently of an assumed positive Weil form. Construction of the model may be an explicit hypothesis; the identity (3.1) must be derived from the specified data.

One usual form is $\Phi_{M,L}(f) = \mathcal{O}_{M,L}(f)\Omega_M$. The observable $\mathcal{O}_{M,L}(f)$ need not be a positive operator. The positivity argument is

$$\langle \mathcal{O}(f)\Omega, \mathcal{O}(g)\Omega \rangle, \quad \|\mathcal{O}(f)\Omega\|^2 \geq 0, \quad (3.2)$$

provided these vectors exist in the positive physical Hilbert space. Writing this formally as an expectation of $\mathcal{O}(f)^\dagger \mathcal{O}(g)$ is justified only on its operator or quadratic-form domain. Linearity of the state map supplies quadratic dependence on f ; the interaction strength of the theory does not affect this argument.

A Euclidean formulation may use a preparation $A_L(f)$ at positive Euclidean time and reflection Θ :

$$B_{M,L}(f, g) = \langle (\Theta A_L(f)) A_L(g) \rangle_M. \quad (3.3)$$

Reflection positivity supplies its positive norm interpretation; a full relativistic reconstruction requires the other Osterwalder–Schrader hypotheses [10]. The arithmetic variable x in (2.9) is not automatically Euclidean time.

An alternative classical construction is a nonnegative joint energy $\mathcal{E}_M(\Phi)$ with boundary trace f , for which $Q_L[f] = \inf_{\text{Tr } \Phi=f} \mathcal{E}_M(\Phi)$. Positivity only in the auxiliary Hessian is insufficient: $w^2 - 2fw$ is coercive in w and has minimum $-f^2$. For nonlinear energies, the assertion that the entire minimized boundary dependence is quadratic is itself a matching requirement.

Proposition 3.2 (Conditional construction-to-RH implication). *If the construction hypotheses for a specified model imply the positive Hilbert space and domains in Definition 3.1, and the complete matching identity is derived under those hypotheses, then construction of that model with those observables implies RH. It is also sufficient to have independently positive regulated pairings $B_{\Lambda,L}$ such that, for every fixed pair of test functions and L ,*

$$\lim_{\Lambda \rightarrow \infty} B_{\Lambda,L}(f, g) = Q_L(f, g). \quad (3.4)$$

Proof. The first assertion follows by taking $g = f$ in (3.1) and applying Theorem 2.1. In the second assertion the finite limit of nonnegative real numbers $B_{\Lambda,L}(f, f)$ is nonnegative. The same criterion applies. No rate uniform in all inputs is required. $\square$

The proposition is a logical consequence of the classical criterion, not the missing reduction theorem. We have not specified a model for which its full premise is verified. An additive subtraction from a regulated norm need not remain positive. Likewise, defining a covariance to equal Q_L , or setting $\Phi = Q_L^{1/2}$ before positivity is known, would reinsert the original problem into the construction hypothesis.

A common source rule compatible under enlargement of I_L is a natural requirement for one field-theoretic realization. Separate positive realizations for every L would already suffice logically for RH; their compatibility is additional structure, not an extra premise of the Weil criterion. A mass gap is not needed for the elementary norm implication. It must be imposed only if a chosen construction or observable limit requires it. An extension of pure Yang–Mills by arithmetic defects or extra fields would create construction obligations beyond the pure Yang–Mills existence problem [12].

3.2 Protected pairings and their adjoints

Sphere and Schur quantization motivate Hilbert completions of protected algebras. The basic data can be described as a complex algebra $\mathcal{A}$, an antilinear automorphism ρ , and a twisted trace T with positive Hermitian form

$$H(a, b) = T(\rho(a)b), \quad T(ab) = T(\rho^2(b)a). \quad (3.5)$$

One quotients null vectors when necessary before completion. On the algebraic core the right and left actions satisfy $R_a^\dagger = L_{\rho(a)}$, since $H(ba, c) = T(\rho(b)\rho(a)c) = H(b, \rho(a)c)$. This does not identify ρ with an arbitrary ordinary adjoint on a single copy of the algebra, nor establish every unbounded closure. The representation formulas used below come from [5, 6, 7]. Classification results for related positive forms give independent mathematical support in specified algebraic families [8, 9]; they do not construct every correlator of an interacting continuum field theory.

The intended conditional program allows an open but credible construction problem. Explicit positive representations, stable regulated models, protected localization formulas and compatible dual descriptions provide different kinds of evidence. The source domain and matching identity still have to be stated for the particular model. No general Schur construction conjecture is promoted to a theorem in the calculations below.

3.3 A necessary source-regularity condition

Proposition 3.3 (Finite-norm orbit smearing cannot match). *Fix $L > 0$. A source of the form $\Phi(f) = \int_{I_L} f(x)V_x\Omega dx$, with V_x unitary, $\Omega \in \mathcal{H}$, and a well-defined strongly measurable orbit, cannot have norm $Q_L[f]$ for all smooth test functions. The same holds for a finite direct sum of such sources.*

Proof. The estimate $\|\Phi(f)\| \leq \sqrt{L}\|\Omega\|\|f\|$ makes the proposed pairing bounded in the input L^2 norm. Let $0 \neq \chi \in C_c^\infty(I_L)$ and $f_N(x) = \chi(x)e^{iNx}$. Equation (2.10) and rapid decay of $\hat{\chi}$ give

$$\frac{Q_L[f_N]}{\log N} \longrightarrow \|\chi\|^2. \quad (3.6)$$

For completeness, after the change $\tau = N + u$ the integrand of the gamma term divided by $\log N$ converges pointwise to $|\widehat{\chi}(u)|^2$. The bound $b((N + u)^2) \leq C[1 + \log(2 + N) + \log(2 + |u|)]$ for $N \geq 2$ permits dominated convergence. All other terms in (2.9) are bounded uniformly in N on this interval. This contradicts boundedness of the proposed norm. $\square$

An exact source must be unbounded relative to the input L^2 norm. An infinite family of insertions or a distributional field with finite output on smooth tests is compatible with this requirement. The proposition makes no assertion of global closability on the union of all intervals, and imposes no obstruction on QFT as such.

4 Sphere quantization and the archimedean data

4.1 The real Gaussian boundary module

The real boundary module of the free hypermultiplet provides a Schrödinger representation [5, Sections 7.2–7.3]. We use oscillator units and the explicitly normalized data

$$\mathcal{H}_{\mathbb{R}} = L^2(\mathbb{R}, dy), \quad X = y = X^\dagger, \quad P = \partial_y = -P^\dagger, \quad \Omega(y) = \pi^{-1/4} e^{-y^2/2}. \quad (4.1)$$

The unitary dilation group is $D_r \psi(y) = e^{r/2} \psi(e^r y)$. Direct integration gives

$$\langle \Omega, D_r \Omega \rangle = (\cosh r)^{-1/2}. \quad (4.2)$$

The real, even Gaussian state fixes a particular Mellin weight; no adjustable gamma shift is introduced.

Proposition 4.1 (Gaussian Mellin normalization). *On even states, define $(\mathcal{U}\psi)(u) = \sqrt{2}e^{u/2}\psi(e^u)$ and let $\mathcal{M}$ be its unitary Fourier transform with factor $(2\pi)^{-1/2}$. Then*

$$(\mathcal{M}\Omega)(\tau) = 2^{-3/4-i\tau/2} \pi^{-3/4} \Gamma(\tfrac{1}{4} - i\tau/2), \quad (4.3)$$

$$\left\| \int_{\mathbb{R}} f(r) D_r \Omega \, dr \right\|^2 = \int_{\mathbb{R}} \rho_G(\tau) |\widehat{f}(\tau)|^2 \, d\tau, \quad \rho_G(\tau) = \frac{|\Gamma(\tfrac{1}{4} + i\tau/2)|^2}{2^{3/2} \pi^{3/2}}. \quad (4.4)$$

The density ρ_G integrates to one.

Proof. Evenness makes $\mathcal{U}$ unitary by the substitution $y = e^u$. The transform of the Gaussian reduces to $\sqrt{2}\pi^{-1/4}(2\pi)^{-1/2} \int_0^\infty y^{-1/2-i\tau} e^{-y^2/2} dy$. The substitution $y^2/2 = t$ evaluates this integral and proves (4.3). Dilations become translations in u , so spectral diagonalization proves (4.4); the possible sign of the Fourier argument is immaterial because ρ_G is even. Plancherel and $\|\Omega\| = 1$ give the last assertion. $\square$

This is an exact positive pairing associated with the quarter-shift. It is not Q_L : its kernel (4.2) is smooth and its spectral density decays exponentially, while (2.10) grows logarithmically. It also falls under Proposition 3.3. Thus this natural source is a normalization control rather than the desired full preparation.

4.2 The oscillator tower and the distinction between a phase and a norm

The positive oscillator $H_{\text{osc}} = \frac{1}{2}(-\partial_y^2 + y^2)$ has, on its even subspace, the spectrum $a_n = 2n + 1/2$. Consequently

$$b(\tau^2) = 2 \text{Tr}_{\mathcal{H}_{\text{even}}} \left[H_{\text{osc}}^{-1} \frac{\tau^2}{H_{\text{osc}}^2 + \tau^2} \right]. \quad (4.5)$$

Each fixed- τ trace converges, since its summands are $O(n^{-3})$. An explicit positive source into $\bigoplus_{n \geq 0} L^2(\mathbb{R})$ is

$$(\mathcal{A}_n F)^\sim(\tau) = \sqrt{\frac{2}{a_n}} \frac{i\tau}{a_n + i\tau} \widehat{F}(\tau), \quad \sum_{n \geq 0} \|\mathcal{A}_n F\|^2 = K[F]. \quad (4.6)$$

Equivalently one may use diagonal Hilbert–Schmidt operators in the oscillator basis. The source is finite on the domain of K and is closed there as a map with its graph norm. Equation (4.5) identifies the required masses with a standard Hamiltonian in a named module. It does not derive the resolvent-weighted source (4.6) as a local insertion or a gluing map of that QFT.

The corresponding unit-modulus archimedean phase is

$$S_\infty(\tau) = \pi^{-i\tau} \frac{\Gamma(\frac{1}{4} + i\tau/2)}{\Gamma(\frac{1}{4} - i\tau/2)}, \quad -i\overline{S_\infty} S'_\infty = b(\tau^2) + w_0. \quad (4.7)$$

Differentiating its logarithm proves the second identity. The scalar factor $\pi^{-i\tau}$ is essential for the contact. Although $|S_\infty| = 1$, its phase derivative at zero is $w_0 < 0$. Unitarity of a multiplication operator therefore does not establish positivity of this derivative. Local-factor phases and gamma splitting are classical; their semilocal use is developed, for example, in [4].

4.3 An interacting benchmark: charged states in $T[SU(2)]$

The $T[SU(2)]$ theory is the three-dimensional $U(1)$ theory with two hypermultiplets. Its sphere module admits a representation on twisted half-densities on $\mathbb{CP}^1$ [5, Section 3.2]. For real FI parameter t , use the coordinate-chart data

$$E = \partial_z, \quad \Omega_t(z) = (1 + |z|^2)^{-1+it}, \quad d\mu = d^2z/\pi. \quad (4.8)$$

This vector has unit norm. Multiplication by $\Gamma(1 - it)$ restores the partition-function factor $Z_t = \pi t / \sinh(\pi t)$, understood continuously as $Z_0 = 1$.

Proposition 4.2 (Charged-state Gram matrix). *The states in (4.8) satisfy*

$$E^n \Omega_t = (-1)^n (1 - it)_n \bar{z}^n (1 + |z|^2)^{-1+it-n}, \quad (4.9)$$

$$\langle E^m \Omega_t, E^n \Omega_t \rangle = \delta_{mn} N_n(t), \quad N_n(t) = \frac{(n!)^2}{(2n+1)!} \prod_{k=1}^n (k^2 + t^2). \quad (4.10)$$

Neither the vacuum returns nor the ratios of these norms supply a nonzero geometric repetition law for bare powers of a fixed multiple of E .

Proof. Repeated differentiation proves (4.9). Angular integration makes different powers orthogonal. With $u = |z|^2$, the remaining radial integral is $\int_0^\infty u^n (1+u)^{-2n-2} du = (n!)^2 / (2n+1)!$. Also $|(1-it)_n|^2 = \prod_{k=1}^n (k^2 + t^2)$. Thus $\langle \Omega_t, E^n \Omega_t \rangle = 0$ for $n > 0$, whereas

$$\frac{N_{n+1}(t)}{N_n(t)} = \frac{(n+1)^2((n+1)^2 + t^2)}{(2n+3)(2n+2)}. \quad (4.11)$$

The ratio at $n = 1$ exceeds that at $n = 0$ by $(19 + t^2)/30 > 0$. A nonzero constant rescaling of E multiplies both ratios by the same positive number and cannot make them equal. $\square$

These are norms in an interacting protected sector. The result excludes the proposed bare charged return, not mixed charged and neutral observables, vortex modules or interfaces. A source $\sum_n c_n(f) E^n \Omega_t$ has norm $\sum_n N_n(t) |c_n(f)|^2$ when the sum converges; an arithmetic application still needs an independent rule for the coefficients $c_n(f)$.

5 Schur pairings before electric dressing

5.1 The conformal electric benchmark

Consider the four-dimensional $\mathcal{N} = 2$ $SU(2)$ theory with four hypermultiplets, with $0 < q < 1$ and trivial flavour fugacities. Let

$$(a; Q)_N = \prod_{j=0}^{N-1} (1 - aQ^j), \quad (a; Q)_\infty = \prod_{j \geq 0} (1 - aQ^j). \quad (5.1)$$

The electric-sector spherical norm from [6, Section 5.1.5] specializes to the finite positive measure

$$\begin{aligned} d\nu_q(\theta) &= Z_q^{-1} \omega_q(\theta) \frac{d\theta}{2\pi}, \quad Z_q = \int_0^{2\pi} \omega_q(\theta) \frac{d\theta}{2\pi}, \\ \omega_q(\theta) &= 4 \sin^2 \theta \frac{|(q^2 e^{2i\theta}; q^2)_\infty|^4}{|(-q e^{i\theta}; q^2)_\infty|^{16}}. \end{aligned} \quad (5.2)$$

The Wilson operator acts as $W = 2 \cos \theta$. Normalizing by Z_q sets the vacuum norm to one; the overall physical normalization would also have to be fixed by a full matching theorem.

For the natural source $a_f(W) = \int f(x) e^{ixW} dx$, the entire pairing is

$$B_q(f, g) = \int_0^{2\pi} \overline{\widehat{f}(-2 \cos \theta)} \widehat{g}(-2 \cos \theta) d\nu_q(\theta). \quad (5.3)$$

Its kernel $\int e^{2ir \cos \theta} d\nu_q(\theta)$ is entire in r , because the spectral variable lies in $[-2, 2]$. It cannot produce separated prime atoms, and its bounded input norm contradicts Proposition 3.3. This conclusion concerns this electric preparation only.

5.2 Local difference factors and the full spherical vector

The elementary factor

$$\chi_q(v) = (-qv; q^2)_\infty^{-1} \quad \text{satisfies} \quad \frac{\chi_q(q^2 v)}{\chi_q(v)} = 1 + qv. \quad (5.4)$$

It occurs in the elementary q -Weyl example of Ambrosino–Gaiotto [7, Section 3.1.1]. The ratio follows by cancelling the products in (5.1). For a trial substitution $q = p^{-1/2}$ and $v = -e^{i\tau \log p}$, let $J_p(\tau) = 1 - p^{-1/2} e^{i\tau \log p}$. Then

$$2 \operatorname{Im} \partial_\tau \log J_p(\tau) = -2(\log p) \sum_{m \geq 1} p^{-m/2} \cos(m\tau \log p). \quad (5.5)$$

The series converges absolutely. The unshifted logarithm of χ_q instead has coefficients with a factor $(1 - q^{2m})^{-1}$. Thus the ratio in (5.4), rather than the factor itself, supplies the geometric coefficient.

This calculation has two immediate limitations. A phase derivative is not a state norm. In addition, varying q with p varies the quantization parameter, and is not a construction in one fixed theory. The dressed source in the next section addresses both points for the positive local reference. In the full massless four-flavour benchmark, the spherical vector has instead the complete ratio

$$\frac{\varphi_0(q^2 v)}{\varphi_0(v)} = \frac{(1 + qv)^8}{(1 - v^2)(1 - q^2 v^2)^2(1 - q^4 v^2)}. \quad (5.6)$$

All matter and vector factors are retained here. Extracting one numerator factor would change the model calculation. Complex holonomy shifts and changes of magnetic charge also require an explicit source map before they can represent translation in the real arithmetic coordinate.

5.3 The common-state RG formulation and the undressed test

The RG formulation [7, Sections 1–2] transports a UV line a to a framed expansion F_a acting on a common spectrum-generator state $\mathcal{S}$. For charge lattice Γ , the relevant quantum torus and pairing take the form

$$X_\gamma X_\eta = q^{\langle \gamma, \eta \rangle} X_{\gamma + \eta}, \quad I(a, b) = c_q \langle F_a \mathcal{S}, F_b \mathcal{S} \rangle_{\ell^2(\Gamma)}, \quad c_q = (q^2; q^2)_\infty^{2r_g} > 0. \quad (5.7)$$

Here $\langle \gamma, \eta \rangle$ is the integral charge pairing and r_g is the gauge rank. The same state enters both the dualization rule and the norm. General domain and construction claims in this formalism include conjectural input; only the explicitly defined elementary Hilbert-space calculations below are used as unconditional statements. An isometric RG description preserves the norm of a given source and cannot itself repair a mismatch.

On one charge ray, take the bilateral shift $Xe_n = e_{n+1}$ on $\ell^2(\mathbb{Z})$ and the vector

$$\mathcal{S} = \sum_{n \geq 0} c_n e_n, \quad c_n = \frac{(-q)^n}{(q^2; q^2)_n}, \quad Z = \sum_{n \geq 0} |c_n|^2 < \infty. \quad (5.8)$$

These are the Fourier coefficients of χ_q . The recurrence $(1 - q^{2n})c_n = -qc_{n-1}$ gives the complete norm identity

$$\|(1 + qX)\mathcal{S}\|^2 = (1 + q^2)Z + 2q \operatorname{Re} \langle \mathcal{S}, X\mathcal{S} \rangle = \sum_{n \geq 0} q^{4n} |c_n|^2. \quad (5.9)$$

Indeed each coefficient of $(1 + qX)\mathcal{S}$ is $q^{2n}c_n$, including the coefficient at $n = 0$.

The actual normalized overlaps do not give q^m . For $m \geq 1$,

$$(-1)^m \frac{\langle \mathcal{S}, X^m \mathcal{S} \rangle}{Z} = \frac{q^m}{Z} \sum_{n \geq 0} \frac{|c_n|^2}{\prod_{j=1}^m (1 - q^{2(n+j)})} > q^m. \quad (5.10)$$

Every summand is positive and each denominator is strictly less than one. Convergence follows from $(q^2; q^2)_n \rightarrow (q^2; q^2)_\infty > 0$. Thus a correct local difference equation does not by itself determine the desired physical return coefficient. This exact failure motivates changing the prepared state while retaining the positive Hilbert metric.

6 Dressed Schur returns and the exact positive prime norm

6.1 The elementary representation and electric preparation

Fix one real parameter $0 < q < 1$. This parameter will be the same for all primes. The elementary q -Weyl representation associated with SQED with one hypermultiplet has the following data [7, Section 3.1.1]. Let

$$\mathcal{H} = \ell^2(\mathbb{Z}) \otimes L^2(S^1, d\theta/(2\pi)), \quad \zeta = e^{i\theta}, \quad (6.1)$$

and write the magnetic charge as m . On the appropriate algebraic line-state domain the actions are

$$\begin{aligned} (v\psi)_m(\zeta) &= q^{-m}\zeta\psi_m(\zeta), \\ (u_-\psi)_m(\zeta) &= (1 + q^{-m-1}\zeta)\psi_{m+1}(q^{-1}\zeta), \\ (u_+\psi)_m(\zeta) &= \psi_{m-1}(q\zeta), \\ (\Omega_q)_m(\zeta) &= C_q\delta_{m0}\chi_q(\zeta), \quad C_q = (q^2; q^2)_\infty. \end{aligned} \quad (6.2)$$

The physical adjoints are inherited from the doubled representation; we do not use $u_- = u_+^\dagger$. Positivity in the calculation is the ordinary positive norm of (6.1). The elementary representation is a protected-sector control, not an identification with the conformal four-flavour theory of Section 5.

For $0 < a < 1$ define, on the zero-charge sector,

$$D_{q,a}(\zeta) = \frac{\sqrt{1-a^2}(-q\zeta; q^2)_\infty}{C_q(1+a\zeta)}. \quad (6.3)$$

The numerator is entire in ζ and the expression is analytic in a neighbourhood of the closed unit disk. Thus its Taylor series defines an electric-line completion, bounded on that sector. It is not asserted to be bounded on all magnetic sectors. The prepared state is explicitly

$$\eta_a = D_{q,a}(v)\Omega_q = \delta_{m0}\frac{\sqrt{1-a^2}}{1+a\zeta}, \quad \|\eta_a\| = 1. \quad (6.4)$$

Multiplication by $-v$ on this sector gives

$$\langle \eta_a, (-v)^k \eta_a \rangle = a^{|k|}, \quad k \in \mathbb{Z}. \quad (6.5)$$

Both assertions follow from the geometric coefficients $\sqrt{1-a^2}(-a)^n$, $n \geq 0$. Thus the descendant factors in (5.10) have been removed by changing the source, without changing the metric. The parameter a is independent of q . This infinite electric preparation is additional specified observable data; it has not been derived as a unique UV boundary condition.

6.2 One magnetic insertion and its graph domain

Lemma 6.1. *If $a = qr$ with $0 < r < 1$, the vector η_{qr} belongs to the graph closure of the algebraic negative magnetic action on its electric-line approximants. Its image is*

$$(u_-\eta_{qr})_m(\zeta) = \delta_{m,-1}\sqrt{1-q^2r^2}\frac{1+\zeta}{1+r\zeta}, \quad \|u_-\eta_{qr}\|^2 = \frac{2(1-q^2r^2)}{1+r}. \quad (6.6)$$

Every closed physical extension of that algebraic action contains this graph limit and has the displayed value.

Proof. The formula for the state follows by substituting (6.4) in (6.2). To justify that substitution as an operator action, let D_N be the Taylor polynomials of $D_{q,qr}$. Its nearest possible pole has modulus $(qr)^{-1} > q^{-1}$, so choose R strictly between these two radii. The polynomials converge uniformly on $|\zeta| \leq R$. Hence $D_N(v)\Omega_q \rightarrow \eta_{qr}$ in norm. On the output sector,

$$u_- D_N(v)\Omega_q = C_q \delta_{m,-1} D_N(q^{-1}\zeta) \chi_q(q\zeta), \quad (6.7)$$

using $(1 + \zeta)\chi_q(q^{-1}\zeta) = \chi_q(q\zeta)$. This converges in norm to the displayed state. In particular no action is being defined by crossing a pole and choosing a boundary value afterwards. Finally,

$$\frac{1 + \zeta}{1 + r\zeta} = 1 + \sum_{n \geq 1} (1 - r)(-r)^{n-1} \zeta^n \quad (6.8)$$

has squared norm $1 + (1 - r)^2/(1 - r^2) = 2/(1 + r)$. $\square$

6.3 The full norm, including its contact

For $d > 0$, set

$$\beta_{q;r,d}^2 = \frac{dr(1+r)}{(1-r)(1-q^2r^2)}, \quad \Phi_{q;r,d}[h] = \beta_{q;r,d} h(-q^{-1}v) u_- D_{q,qr}(v) \Omega_q \quad (6.9)$$

for a finite Laurent polynomial h . On the output sector $m = -1$, $v = q\zeta$, so $-q^{-1}v = -\zeta$. This normalization of the electric argument is part of the source prescription.

Theorem 6.2 (Schur realization of the positive local prime reference). *With $z = -\zeta$, the preparation (6.9) satisfies*

$$\langle \Phi_{q;r,d}[h], \Phi_{q;r,d}[k] \rangle = \int_{S^1} \overline{h(z)} k(z) w_{r,d}(z) \frac{d\theta}{2\pi}, \quad (6.10)$$

$$w_{r,d}(z) = \frac{dr(1+r)}{1-r} \frac{|1-z|^2}{|1-rz|^2} = \kappa_{r,d} - d \sum_{n \geq 1} r^n (z^n + z^{-n}), \quad (6.11)$$

$$\kappa_{r,d} = \frac{2dr}{1-r}. \quad (6.12)$$

The series converges absolutely and uniformly. In the Laurent basis,

$$\langle \Phi[z^j], \Phi[z^k] \rangle = \begin{cases} \kappa_{r,d}, & j = k, \\ -dr^{|j-k|}, & j \neq k. \end{cases} \quad (6.13)$$

For each prime p , setting $r = p^{-1/2}$ and $d = \log p$ supplies every single-prime off-diagonal coefficient in the Weil form. The same fixed q works for all these preparations.

Proof. Lemma 6.1 and (6.9) give the first expression for $w_{r,d}$ directly from the physical norm. To expand it, use the Poisson-kernel identity

$$\frac{1 - r^2}{|1 - rz|^2} = 1 + \sum_{n \geq 1} r^n (z^n + z^{-n}) \quad (|z| = 1). \quad (6.14)$$

Subtracting this identity from $(1+r)/(1-r)$ and multiplying by d gives (6.11). Integration against $\overline{z^j} z^k$ gives (6.13). Finally $a_p = q/\sqrt{p} < q$, so the graph-domain proof applies to every prime at the same q . All operators needed after the magnetic action are bounded on its fixed output sector for Laurent polynomial inputs. $\square$

The source normalization β uses the assigned arithmetic boundary data. It determines the entire multiplier; no independent contact is subtracted. Since $w_{r,d}$ is bounded, (6.10) extends by density from Laurent polynomials to $L^2(S^1)$ as a bounded source map into the output sector.

Corollary 6.3 (Sharp contact in this stationary scalar class). *A continuous scalar circle multiplier with nonzero Fourier coefficients $-dr^{|n|}$ can be nonnegative only if its constant coefficient is at least $\kappa_{r,d}$. The multiplier in (6.11) attains that value.*

Proof. The multiplier with constant coefficient C equals $C - d \sum_{n \geq 1} r^n (z^n + z^{-n})$. Its value at $z = 1$ is $C - 2dr/(1 - r)$, which must be nonnegative. Equation (6.11) is nonnegative and vanishes at $z = 1$. $\square$

The same statement holds for the corresponding whole-line stationary multiplier. It is not a claim of optimality over all finite-interval or nonstationary realizations. This is the known positive local reference; the result here supplies its Schur operator factor and checked domain.

6.4 An explicit lift to continuous arithmetic inputs

The circle coordinate is not, by itself, the real logarithmic input. To give an actual source map, define

$$(\mathcal{B}_d F)_j(\theta) = d^{-1/2} \hat{F}((\theta + 2\pi j)/d), \quad 0 \leq \theta < 2\pi, \quad j \in \mathbb{Z}. \quad (6.15)$$

This is a unitary map from $L^2(\mathbb{R})$ to $L^2(S^1, d\theta/(2\pi); \ell^2(\mathbb{Z}))$. Indeed the intervals $[2\pi j/d, 2\pi(j+1)/d]$ partition the Fourier line, and (2.1) gives equality of norms. Conversely an arbitrary square-integrable family specifies a Fourier transform interval by interval, giving surjectivity.

Apply the extended source of Theorem 6.2 to each component of $\mathcal{B}_d F$ in an orthogonal copy of $\mathcal{H}$. The resulting positive norm is

$$\begin{aligned} B_{r,d}[F] &= \frac{1}{2\pi} \int_{\mathbb{R}} w_{r,d}(e^{id\tau}) |\hat{F}(\tau)|^2 d\tau \\ &= \kappa_{r,d} \|F\|^2 - 2d \sum_{n \geq 1} r^n \operatorname{Re} \langle F, U_{nd} F \rangle. \end{aligned} \quad (6.16)$$

The Fourier sign for U_d does not change the expression, because both orientations occur. For $F = E_L f$, only $nd < L$ remains.

This construction adds a specified multiplicity space and preparation. That multiplicity is not identified with the magnetic-charge label already present in (6.1). No particular four-dimensional interface has yet been shown to implement (6.15). The result is an exact positive Hilbert-space realization containing an elementary Schur operator factor. A full field-theoretic interpretation of the lift is part of the open matching problem.

7 Magnetic domains and joint-prime gluing

7.1 What higher magnetic powers would require

The domain statement for one insertion is sufficient for Theorem 6.2. It does not place the prepared state in the domain of every negative magnetic power.

Proposition 7.1 (Formal higher-power coefficient criterion). *For $k \geq 1$, iteration of the coefficient action gives the formal series*

$$(u_-^k \eta_a)_{-k}(\zeta) = \sqrt{1 - a^2} \frac{\prod_{j=0}^{k-1} (1 + q^{k-1-2j} \zeta)}{1 + a q^{-k} \zeta}. \quad (7.1)$$

Its Taylor coefficient sequence belongs to ℓ^2 if and only if

$$a < q^k \quad \text{or} \quad a \in \{q, q^3, \dots, q^{2k-1}\}. \quad (7.2)$$

For fixed $0 < a < 1$, square summability of all these formal outputs is equivalent to $a = q^{2j+1}$ for some integer $j \geq 0$.

Proof. Induction in (6.2) gives (7.1). For every coefficient of degree $n \geq k$, expanding the denominator shows that the coefficient equals

$$\sqrt{1-a^2}(-aq^{-k})^n \prod_{j=0}^{k-1} (1 - q^{2j+1}/a). \quad (7.3)$$

If the product is nonzero, the geometric tail is square summable exactly when $aq^{-k} < 1$. If the product vanishes, the numerator cancels the denominator and the output is a polynomial. This proves (7.2). For all k , the inequality eventually fails at any fixed $a > 0$, so an exceptional value is necessary. If $a = q^{2j+1}$, it satisfies the strict inequality for $k \leq j$ and the cancellation condition for $k \geq j+1$, proving sufficiency for the formal coefficient statement. $\square$

The exceptional square-summable outputs do not by themselves prove membership in powers of a chosen closed operator extension. A divergent coefficient tail is, however, a necessary domain obstruction whenever the prescribed action on Fourier coefficients is retained. This distinction prevents meromorphic continuation across a pole from being mistaken for an operator-domain argument.

For the prime states $a_p = qp^{-1/2}$, requiring all negative powers would force $p^{-1/2} = q^{2j_p}$. This cannot hold simultaneously for $p = 2$ and $p = 3$, since it would imply a nontrivial equality between an integer power of 2 and an integer power of 3. It does not affect the single insertion already proved. More generally, for any fixed finite $K > 1$, choosing $q^{K-1} > 2^{-1/2}$ puts every prime state in the strictly decaying regime for all powers through K . In that regime the Taylor-approximation argument of Lemma 6.1 extends through the finite sequence of shifts. A proposed junction must specify the finite words or actual operator domains it uses; a field theory need not put every state in the domain of every unbounded power.

7.2 A two-prime obstruction for fixed output vectors

For each prime write $r_p = p^{-1/2}$, $d_p = \log p$, and

$$A_p(z) = \frac{1-z}{1-r_p z} = 1 - (1-r_p) \sum_{n \geq 1} r_p^{n-1} z^n, \quad c_p = \frac{d_p r_p (1+r_p)}{1-r_p}. \quad (7.4)$$

The norm of $\sqrt{c_p} A_p$ is the multiplier (6.11). Consider a proposed coherent sum with fixed unit output vectors v_2, v_3 :

$$F \mapsto \hat{F}(\tau) \left[\sqrt{c_2} A_2(e^{id_2 \tau}) v_2 + \sqrt{c_3} A_3(e^{id_3 \tau}) v_3 \right], \quad (7.5)$$

with output measure $d\tau/(2\pi)$.

Proposition 7.2 (Mixed atom from constant-vector gluing). *In (7.5), the coefficient of $e^{i(\log 3 - \log 2)\tau}$ in the full norm multiplier is*

$$\sqrt{c_2 c_3} (1-r_2)(1-r_3) \langle v_2, v_3 \rangle. \quad (7.6)$$

Consequently this source, together with independently added gamma, contact and pole terms, cannot equal the two-prime Weil form on $I_{5/4}$ unless $v_2 \perp v_3$. In the orthogonal case its prime part is exactly the sum of the separate positive references.

Proof. The first cross term in the squared norm is $\sqrt{c_2 c_3} \overline{A_2(e^{id_2 \tau})} A_3(e^{id_3 \tau}) \langle v_2, v_3 \rangle$. Its first non-constant coefficient from each factor gives (7.6). A second pair of nonnegative indices with $n \log 3 - m \log 2 = \log 3 - \log 2$ would give $3^{n-1} = 2^{m-1}$ and hence $n = m = 1$. The reverse cross term cannot give this positive frequency with nonnegative indices, and the individual squared norms contain only integer multiples of d_2 or d_3 .

The Fourier coefficient series of each factor is absolutely summable; the cross kernel is therefore a finite-variation atomic measure. Other mixed lengths may accumulate, but its atom at $\log(3/2)$ has the specified nonzero mass if the overlap is nonzero. The Weil target has no prime-power atom there. Its gamma and pole kernels are smooth near this nonzero separation, and its contact is supported at zero, so none can cancel the atom. Since $0 < \log(3/2) < 5/4$, restriction to $I_{5/4}$ still detects it. Equality of all polarized test-function pairings would imply equality of their distribution kernels, a contradiction. A purely imaginary overlap gives an imaginary antisymmetric kernel; complex tests still detect it. Thus the overlap must vanish. $\square$

The proposition does not apply to arbitrary coherent gluing. A joint operator-valued junction may have additional terms which cancel mixed returns as an identity. Producing and evaluating such a junction is a new source construction. The failed fixed-vector ansatz gives no reason to estimate its unwanted atoms as a small residual.

7.3 The full comparison, with no omitted contact or pole

Let $\mathcal{P}$ be any finite set containing all primes $p < e^L$. Equations (2.9) and (6.16) give the exact identity

$$\begin{aligned} Q_L[f] &= K[E_L f] + \sum_{p \in \mathcal{P}} B_{r_p, d_p}[E_L f] \\ &\quad + \left(w_0 - \sum_{p \in \mathcal{P}} \kappa_{r_p, d_p} \right) \|f\|^2 + P_L[f]. \end{aligned} \tag{7.7}$$

Each term in the first line is independently positive. The second line has not been realized as part of their joint norm. Extra primes in $\mathcal{P}$ contribute only contacts on this interval, which cancel algebraically against the displayed subtraction. This does not make the subtraction an independently positive operation.

Thus the dressed result arrives at the same contact and pole problem as the conservative prime reference. It supplies a concrete Schur preparation and removes the earlier undressed-state mismatch, but establishes no larger positivity interval. Unitary repackaging of the orthogonal gamma and prime outputs leaves (7.7) unchanged. The next calculation must determine contact, poles and prime interference within one complete positive preparation.

8 The next complete matching problem

8.1 A finite-prime phase identity is useful background

For comparison with earlier local-factor approaches, define $S_p = J_p / \overline{J_p}$ using (5.5), and set $S_{\mathcal{P}} = S_{\infty} \prod_{p \in \mathcal{P}} S_p$ for a finite prime set. Logarithmic differentiation gives

$$-i \overline{S_{\mathcal{P}}} S'_{\mathcal{P}} = b(\tau^2) + w_0 - 2 \sum_{p \in \mathcal{P}, m \geq 1} (\log p) p^{-m/2} \cos(m\tau \log p). \tag{8.1}$$

When $\mathcal{P}$ contains the active primes of I_L , integration against $|\widehat{E_L f}|^2/(2\pi)$ and addition of $P_L[f]$ gives $Q_L[f]$. Inactive repetitions vanish by support. This argument uses no infinite Euler product on the critical line.

The identity is classical local-factor algebra. Its phase derivative is negative at $\tau = 0$, and the signed poles are outside it. It is neither a positive norm nor an identification of a canonical semilocal positive pairing with the Weil form. The present inverse problem is to find a physical source for the whole form, not merely for this phase.

8.2 A precise open problem

Problem 8.1 (Joint arithmetic preparation). Specify a positive quantum, boundary or defect model, with a physically defined source $\Phi_L(f)$ on smooth compact support, for which the entire pairing is (2.9). Derive the source's adjoint, domain and normalization, including the fixed contact, both signed poles, every active prime repetition, and absence or exact cancellation of mixed translations. For an all-support realization, give one compatible rule as L increases.

This is an open problem, not an existence conjecture established for a named field theory. A worthwhile intermediate theorem could be conditional on construction of a credible specified theory, provided the construction assumptions do not prescribe the desired covariance. Proposition 3.2 would then give the forward implication to RH. Neither a reverse implication nor an equivalence with Yang–Mills is proposed here.

The two leading directions have complementary roles. Sphere boundary modules give the clearest real Gaussian and oscillator normalization control. The next sphere object should be a boundary or vortex correspondence acting on that control with a derived infinite source. An arbitrary choice of coefficients in an orthogonal basis would not establish the required physical correspondence. The Schur direction now has a specified positive prime source, and its next object should be a charge-neutral gauge or RG/interface preparation.

In the conformal $SU(2)$ benchmark, the magnetic generator has two shifted contributions and a neutral coefficient, schematically

$$H = uC_+(v)u + C_0(v) + u^{-1}C_-(v)u^{-1}, \quad (8.2)$$

and the representation includes a Weyl constraint [6, Sections 5.1.2–5.1.3]. The notation in (8.2) describes the structure to retain; it is not yet a specified arithmetic source. The exact coefficients and projection must be carried through a chosen neutral line or interface word. The neutral term and projection are possible mechanisms for joint interference, not terms whose desired effect can be assumed. This calculation must also implement the real-input lift or replace it by another derived boundary map.

The immediate target is the complete identity at $L = 1$:

$$\|\Phi_1(f)\|^2 = K[E_1 f] + w_0\|f\|^2 + P_1[f] - \frac{2\log 2}{\sqrt{2}} \operatorname{Re}\langle E_1 f, U_{\log 2} E_1 f \rangle. \quad (8.3)$$

The gamma response, contact, pole terms and prime atom must be computed together. At $L = 5/4$ the atom at $\log(3/2)$ must be absent; at $L = 3/2$ the second return of 2 must have primitive coefficient $(\log 2)/2$ for each oriented translation, while the prime 3 term remains active. These are complete identity tests. Failure calls for a different source or model rather than a growing sequence of residual estimates.

8.3 Scope of the working manuscript

The positive gamma action, its integer refinement, and the positive prime reference are established ingredients. The new dressed calculation places the last of these in an explicit elementary Schur preparation. The source tests in this manuscript are sufficiently explicit to be checked and reused, but they have not supplied the full matching identity. The main remaining difficulty has therefore not been reduced to a numerical sign check.

Other constructions remain available: interacting boundary gluing, integrable defects, gauge networks and arithmetic graph models may provide different source rules. Their broader literature comparison is retained in the investigation's research notes, while this working manuscript focuses on the calculations above. The evidence for sphere and Schur systems is the availability of explicit protected representations and positive pairings, not an assertion that their arithmetic matching is inevitable. Further drafts should strengthen a specific joint construction or replace it when its complete pairing fails.

A A positive gamma action and its integer refinement

This appendix records the common positive bulk from which a different joint source could be built. For $\alpha > 0$, let

$$a_n(\alpha) = 2(n + \alpha), \quad b_\alpha(s) = \sum_{n \geq 0} \frac{2}{a_n(\alpha)} \frac{s}{a_n(\alpha)^2 + s}. \quad (\text{A.1})$$

Thus $b = b_{1/4}$. For line fields u_n with weak derivatives, define the nonnegative joint energy

$$\mathcal{E}_\alpha(F, \{u_n\}) = \sum_{n \geq 0} \frac{2}{a_n(\alpha)} \left(\|F - u_n\|^2 + a_n(\alpha)^{-2} \|u'_n\|^2 \right). \quad (\text{A.2})$$

The finite-energy domain is specified by the sum as a whole. In particular, the divergent expression $\sum_n 2\|F\|^2/a_n(\alpha)$ must not be split off as a separate energy from compensating fields.

Proposition A.1 (Elimination and positive refinement). *For smooth compactly supported F ,*

$$\inf_{\{u_n\}} \mathcal{E}_\alpha(F, \{u_n\}) = \frac{1}{2\pi} \int_{\mathbb{R}} b_\alpha(\tau^2) |\widehat{F}(\tau)|^2 d\tau. \quad (\text{A.3})$$

For each integer $p \geq 2$, put $\alpha_j = (\alpha + j)/p$, $0 \leq j < p$, and $D_p F(x) = p^{-1/2} F(x/p)$. Then

$$b_\alpha(s) = \frac{1}{p} \sum_{j=0}^{p-1} b_{\alpha_j}(s/p^2), \quad (\text{A.4})$$

$$\mathcal{E}_\alpha(F, \{u_n\}) = \frac{1}{p} \sum_{j=0}^{p-1} \mathcal{E}_{\alpha_j}(D_p F, \{D_p u_{pm+j}\}_{m \geq 0}). \quad (\text{A.5})$$

The latter identity is valid as an equality of nonnegative extended energies, before eliminating the fields.

Proof. For each mode, minimizing the Fourier-space quadratic function gives $\widehat{u}_n = a_n^2(a_n^2 + \tau^2)^{-1} \widehat{F}$ and the summand in (A.1). Substitution of all minimizers, or finite towers followed by monotone convergence, proves (A.3). For refinement, write $n = pm + j$. Then $a_{pm+j}(\alpha) = p a_m(\alpha_j)$, $\|D_p u\| = \|u\|$, and $\|(D_p u)'\| = p^{-1} \|u'\|$. Substitution proves (A.5); reindexing the positive multiplier sum gives (A.4). $\square$

The first examples are

$$\begin{aligned} b_{1/4}(s) &= \tfrac{1}{2}b_{1/8}(s/4) + \tfrac{1}{2}b_{5/8}(s/4), \\ b_{1/4}(s) &= \tfrac{1}{3} \left[b_{1/12}(s/9) + b_{5/12}(s/9) + b_{3/4}(s/9) \right]. \end{aligned} \quad (\text{A.6})$$

Refining by p and then by another integer ℓ gives residue labels $j + pk$, the dilation $D_{p\ell}$ and weights $1/(p\ell)$. Thus refinement by 2 then 3, or 3 then 2, gives the same six branches $\{1, 5, 9, 13, 17, 21\}/24$ up to permutation.

There is also a contact identity. Set $m_\alpha(\tau) = \text{Re } \psi(\alpha + i\tau/2) - \log \pi$. Gauss multiplication [3] gives

$$m_\alpha(\tau) = \log p + \frac{1}{p} \sum_{j=0}^{p-1} m_{\alpha_j}(\tau/p). \quad (\text{A.7})$$

Subtracting the value at zero recovers (A.4). Only the kinetic actions in (A.2) have the asserted positivity; (A.7) is not a positive decomposition of the complete archimedean form.

These formulas exist for every integer, and so do not select primes. Coordinate dilation by p is also different from translation by $\log p$. A conservative junction acting on the common refined fields could be tested, but its injection, return and readout maps would have to derive these relations. The normalization $p^{-1/2}$ in the residual-output inclusion is fixed by (A.5); it is not yet a physical arithmetic return amplitude. Refinement or a unitary rotation of all outputs preserves their norm and cannot itself introduce prime terms. No novelty claim is made for gamma multiplication or its semilocal use [4].

B Gauge transfer and two first-prime source tests

B.1 Winding a holonomy is different from transfer evolution

Two-dimensional heat-kernel Yang–Mills supplies a rigorous control with explicit Wilson functionals [11]. Normalize the positive $SU(2)$ Casimir so that the spin- j representation has value $j(j+1)$. For dimensionless area-coupling parameter $s > 0$,

$$\mathbb{E}_s[\chi_j(U)] = (2j+1)e^{-sj(j+1)}. \quad (\text{B.1})$$

The normalized fundamental loop wound m times is $w_m(U) = \frac{1}{2} \text{Tr}_{\mathbb{C}^2}(U^m) = \cos(m\vartheta)$, for eigenvalues $e^{\pm i\vartheta}$. The character identity $2 \cos(m\vartheta) = \chi_{m/2} - \chi_{(m-2)/2}$ for $m \geq 2$ gives

$$\mathbb{E}_s[w_1] = e^{-3s/4}, \quad \mathbb{E}_s[w_m] = \frac{1}{2} \left[(m+1)e^{-sm(m+2)/4} - (m-1)e^{-sm(m-2)/4} \right]. \quad (\text{B.2})$$

If the first return is set to $r \in (0, 1)$, the second is $(3r^{8/3} - 1)/2$, strictly less than r^2 . Indeed, for $y = r^{2/3}$,

$$3y^4 - 2y^3 - 1 = (y-1)(3y^3 + y^2 + y + 1) < 0. \quad (\text{B.3})$$

No overall primitive-length coefficient repairs this relative failure.

The same literal winding rule fails for every central probability convolution semigroup with proposed moments $\mathbb{E}_t[w_1] = r_t = e^{-\lambda t}$ and $\mathbb{E}_t[w_2] = r_t^2$, $\lambda > 0$. The normalized spin-one character coefficient would be $c_1(t) = (1 + 2r_t^2)/3$. Convolution requires multiplicativity, but

$$c_1(t+u) - c_1(t)c_1(u) = \frac{2}{9}(1 - r_t^2)(1 - r_u^2) > 0 \quad (t, u > 0). \quad (\text{B.4})$$

In contrast, let $H = \frac{2}{3}C_2$ on the character Hilbert space with Haar inner product, and choose the unit vector $\eta = \chi_{1/2}$. Then $H\eta = \frac{1}{2}\eta$ and

$$\langle \eta, e^{-m(\log p)H} \eta \rangle = p^{-m/2}. \quad (\text{B.5})$$

This propagates a fixed representation for a longer interval; it does not wind the same holonomy. It passes the scalar repetition test.

Even so, the literal affine Euclidean source $\Phi_L(f) = \int_{I_L} f(x) e^{-(b+x)H} \eta dx$, $b > L/2$, has a Hankel kernel $C(2b + x + y)$, where $C(t) = \langle \eta, e^{-tH} \eta \rangle$. If $H \geq 0$ and this kernel is invariant under small simultaneous translations of x, y , then C' vanishes on an interval. Its spectral measure satisfies $-C'(t) = \int \lambda e^{-t\lambda} d\nu_\eta(\lambda)$, so η must lie in the zero-energy subspace. The resulting kernel is constant. This excludes that affine-time source, not other spatial, normalized or extended gauge preparations.

B.2 A positive disk system for masses and returns

Let Λ be the Dirichlet-to-Neumann operator on the unit disk. On the reflection-even boundary subspace, with normalized measure $d\vartheta/(2\pi)$, use the basis $e_0 = 1$, $e_n = \sqrt{2} \cos(n\vartheta)$ for $n \geq 1$. Harmonic extension gives $\Lambda e_n = n e_n$. Thus

$$A = 2\Lambda + \frac{1}{2}\text{id}, \quad A e_n = a_n e_n, \quad (\text{B.6})$$

has the gamma spectrum and positive quadratic form

$$\langle h, Ah \rangle = \frac{2}{2\pi} \int_{\mathbb{D}} |\nabla u_h|^2 d^2z + \frac{1/2}{2\pi} \int_{S^1} |h|^2 d\vartheta. \quad (\text{B.7})$$

Here u_h is the harmonic extension; the form domain is the even $H^{1/2}$ trace space. The choice of half-unit shift and one mode per nonnegative integer is explicit model data, not forced by an arithmetic symmetry.

The transfer is a shifted Poisson semigroup. On the full circle it also has the positive subordination formula

$$e^{-tA} = e^{-t/2} \frac{t}{\sqrt{\pi}} \int_0^\infty s^{-3/2} e^{-t^2/s} e^{-sD^2} ds, \quad D = -i\partial_\vartheta, \quad (\text{B.8})$$

which restricts to the even subspace. The scalar Laplace integral proves the formula on each Fourier mode. This is a change of transfer generator, not an equivalence with the original Yang–Mills dynamics.

Let $P_0 = |e_0\rangle\langle e_0|$ and $T_p = e^{-(\log p)A}$. Then

$$b(\tau^2) = 2 \text{Tr} \left[A^{-1} \frac{\tau^2}{A^2 + \tau^2} \right],$$

$$\text{Tr}(P_0 T_p^m) = p^{-m/2}, \quad T_p T_\ell = T_{p\ell}, \quad \text{Tr}(T_p^m) = \frac{p^{-m/2}}{1 - p^{-2m}}. \quad (\text{B.9})$$

The ground projection is therefore essential to the proposed pure geometric return. This model does not derive primality, the primitive coefficient $\log p$, or the absence of mixed histories.

B.3 An orthogonal first-prime branch has an unavoidable cusp

Use the source (4.6) and set $a = 1/2$, $d = \log 2$, $r = e^{-ad} = 2^{-1/2}$. With $H_x = -\partial_x^2$, consider the full positive preparation

$$F \mapsto \left(\{\mathcal{A}_n F\}_{n \geq 1}, \sqrt{1-\theta} \mathcal{A}_0 F, \sqrt{\theta} \mathcal{A}_0 (\text{id} - r U_d) F \right), \quad 0 \leq \theta \leq 1. \quad (\text{B.10})$$

Its Gram operator is, in form notation, $K + \theta(2/a)H_x(a^2 + H_x)^{-1}[r^2 \text{id} - r(U_d + U_d^*)]$. Matching the prime atom forces $\theta = ad/2 = d/4$. The resulting complete pairing has operator

$$B = K + dr^2 \text{id} - dr(U_d + U_d^*) + R_{a,d}, \quad R_{a,d} = -da^2[r^2 \text{id} - r(U_d + U_d^*)](a^2 + H_x)^{-1}. \quad (\text{B.11})$$

The resolvent kernel $e^{-a|s|}/(2a)$ gives

$$R_{a,d}(s) = \frac{ad}{2} \begin{cases} r^2 e^{a|s|}, & |s| \leq d, \\ e^{-a|s|}, & |s| \geq d. \end{cases} \quad R'_{a,d}(d+) - R'_{a,d}(d-) = -da^2 r \neq 0. \quad (\text{B.12})$$

On $d < L < \log 3$, after the common prime atom is removed, the candidate-minus-target difference is $(dr^2 - w_0) \text{id} + E_L^* R_{a,d} E_L - P_L$. The pole kernel is smooth near $x - y = d$ and the contact has no support there. Neither can cancel the cusp, so the full pairing cannot match on that interval. This is an exact kernel obstruction, requiring no residual bound.

For a finite positive sum of analogous orthogonal branches with mass a_n , positive weight θ_n and attenuation r_n , the derivative jump at d is $-2\theta_n a_n r_n$ for each active branch. The signs agree and cannot cancel. General coherent changes of source are outside this assertion.

B.4 A coherent repair cancels the cusp and still fails the contact

In two common outputs, use the lowest gamma source and a shift source

$$\Gamma_0 F = \begin{pmatrix} 2H_x(a^2 + H_x)^{-1} F \\ \partial_x(a^2 + H_x)^{-1} F \end{pmatrix}, \quad Z_{u,v} F = \begin{pmatrix} u(U_d - U_d^*) F \\ vF + u(U_d + U_d^*) F \end{pmatrix}, \quad a = \frac{1}{2}, \quad (\text{B.13})$$

where u, v are real. All operators commute on the line. The first components are respectively self-adjoint and skew-adjoint, and the second components respectively skew-adjoint and self-adjoint. Hence

$$\Gamma_0^* Z + Z^* \Gamma_0 = 0, \quad Z^* Z = (v^2 + 4u^2) \text{id} + 2uv(U_d + U_d^*). \quad (\text{B.14})$$

Keeping the other gamma modes and setting $2uv = -dr$ gives

$$B_{\text{coh}} = K + C \text{id} - dr(U_d + U_d^*), \quad C = v^2 + 4u^2 \geq 2dr. \quad (\text{B.15})$$

The inequality follows from $C - 2dr = (v + 2u)^2$. There is no displaced continuous term, but on the first-prime interval

$$B_{\text{coh},L} - Q_L = (C - w_0) \text{id} - P_L \neq 0. \quad (\text{B.16})$$

Indeed $C - w_0 > 0$ while P_L has rank two on an infinite-dimensional input space. This coherent representation is still the sum of independent gamma and jump Gram forms, because its Hermitian interference vanishes. It supplies a second exact stopping criterion for the tested source.

C A finite rational-feedback obstruction

A common attempt to fit a negative contact is to filter a positive reference source by finitely many rational functions of its Gram operator. The following elementary argument excludes a specific class without estimating its residual.

Proposition C.1. *Let A_0 be a closed source into a positive Hilbert space and let $T = A_0^* A_0 > 0$ have infinitely many distinct eigenvalues. On finite linear combinations of eigenvectors, form the direct-sum source*

$$A_{\text{rat}} f = (A_0 r_1(T) f, \dots, A_0 r_m(T) f, s_1(T) f, \dots, s_k(T) f), \quad (\text{C.1})$$

where all r_j, s_ℓ are rational functions with complex coefficients, finite at every eigenvalue. For any $\alpha > 0$, this source cannot satisfy

$$\|A_{\text{rat}} f\|^2 = \langle f, (T - \alpha \text{id}) f \rangle \quad (\text{C.2})$$

on all eigenvectors, even if $T - \alpha \text{id}$ is positive on its spectrum. Positive constant metrics within either group of channels can be absorbed into the rational functions and obey the same conclusion.

Proof. For an eigenvalue λ , the proposed identity gives $\lambda \sum_j |r_j(\lambda)|^2 + \sum_\ell |s_\ell(\lambda)|^2 = \lambda - \alpha$. Write $r^\#(z) = \overline{r(\bar{z})}$ and define the rational function

$$R(z) = z \sum_j r_j(z) r_j^\#(z) + \sum_\ell s_\ell(z) s_\ell^\#(z). \quad (\text{C.3})$$

Clearing denominators, infinitely many distinct roots force $R(z) = z - \alpha$ as a rational identity. Choose a real x with $0 < x < \alpha$ outside the finite pole set. Its sum-of-squares representation gives $R(x) \geq 0$, while $x - \alpha < 0$, a contradiction. $\square$

No finite accumulation point of eigenvalues is needed. A nonzero polynomial has only finitely many roots. Finite cascades of scalar rational filters of the same reference remain in this class. For example,

$$\lambda \left(1 - \frac{\kappa}{\lambda + \mu} \right)^2 = \lambda - 2\kappa + \frac{2\kappa\mu}{\lambda + \mu} + \frac{\kappa^2\lambda}{(\lambda + \mu)^2}, \quad (\text{C.4})$$

whose last terms are nonnegative for $\lambda, \kappa > 0, \mu \geq 0$. Adding finitely many channels of (C.1) cannot remove the exact constant in this reference-only class.

The proposition does not exclude general finite-field models, nonrational responses from continuum dynamics, couplings to operators other than functions of T , infinite channel families, cross couplings outside (C.1), or agreement on only finitely many inputs. It is a scoped selection result with no assertion of literature priority.

D Checks, provenance and mathematical status

The analytical arguments in this draft are written proofs and explicitly qualified construction problems. Four small standard-library Python programs accompany the investigation. They check finite algebra and normalization identities rather than positivity of the Weil form. The preserved records distinguish exact rational checks from labelled floating-point illustrations.

Program in <code>numerics/</code>	Cases	Scope
<code>check_matching.py</code>	153	Rational filters and gamma refinement
<code>check_gauge_transfer.py</code>	326	Winding, transfer and first-prime algebra
<code>check_sphere_schur.py</code>	251	Charged norms, product ratios and undressed overlaps
<code>check_dressed_schur.py</code>	587	Dressed norm, magnetic coefficients and mixed-return witnesses

The 1,317 finite cases are supporting checks, not 1,317 independent theorems. Finite-product end factors and truncation boundary coefficients are retained. The finite mixed-return sample does not replace the unique-factorization proof in Proposition 7.2. The Mellin and beta integrals, infinite-series convergence, graph-domain argument and kernel exclusions are justified in the text; the programs do not formally verify them.

The manuscript is self-contained at the level of its stated external inputs: the classical Weil criterion and the cited model representation data. It does not input a shared background file or cite equation numbers from one. The accompanying coverage map relates its sections to the retained research notes, including the broader candidate survey. Build instructions and a hash record identify the reviewed sources and compiled PDF. The `drafts/` folder preserves complete dated source-and-PDF snapshots. Those hashes establish file identity, not mathematical correctness or portability of PDF metadata between TeX installations.

The following boundaries of the results are essential. The positive prime reference is known; the dressed theorem gives an explicit elementary Schur source for it. Square-summable formal magnetic outputs do not settle all closed-operator extensions. The continuous lift has not been derived from a four-dimensional interface. The general protected-sector construction hypotheses have not been proved here. Most importantly, the full contact-and-pole identity in Problem 8.1 remains unsolved, so this draft makes no claim to establish RH or a completed reduction to constructive QFT.

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